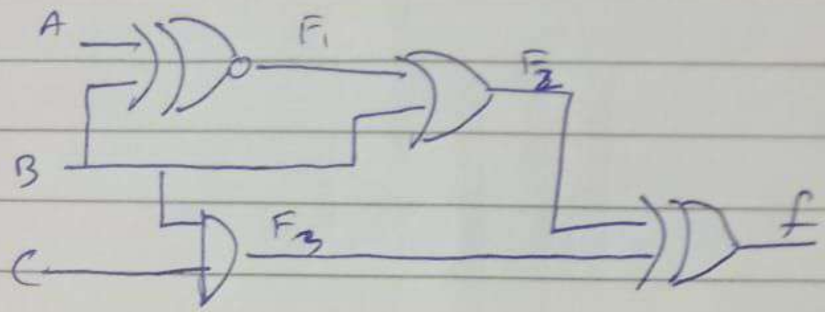
DAY-2 Assignment

1)a)



$F_1 = \bar{A}\bar{B} + AB$

$F_2 = \bar{A}\bar{B} + AB + B$

$= \bar{A}\bar{B} + B \Rightarrow \bar{A} + B$

$F_3 = BC$

$f = \underbrace{(\bar{A} + B)}_{X} \oplus \underbrace{BC}_{Y}$

$= \bar{X}Y + \bar{Y}X$

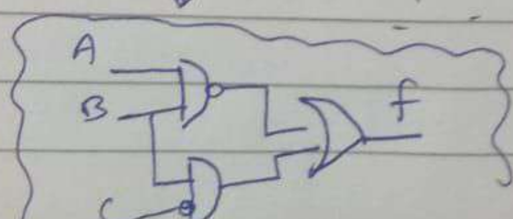
$\Rightarrow \overline{(\bar{A} + B)}(BC) + \bar{A}\bar{B} + \bar{A}\bar{C} + B\bar{C}$

$\underbrace{(A\bar{B})(BC)}_{0} + \bar{A}\bar{B} + \bar{A}\bar{C} + B\bar{C}$

Resultant ↓

$\therefore f = \bar{A}\bar{B} + B\bar{C}$

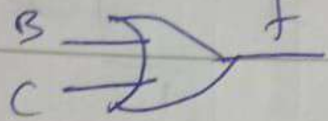
($\because$ Consensus theorem)



$$F_3 = C$$

$$f = B + C$$

Resultant ↓



2)

a) $\bar{A}C\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCE + ABD$

$\cancel{\bar{A}\bar{D}(CE+\bar{B}) + AB(CE+D)}$

$\Rightarrow \bar{A}\bar{D}(\bar{B}+CE) + AB(D+CE)$

b) $\bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D} + A\bar{C}D + A\bar{B}\bar{C}$

$\Rightarrow \bar{B}\bar{C}(A+\bar{A}) + \bar{A}C(1+\bar{D}) + AD(B+\bar{C})$

$\Rightarrow \bar{B}\bar{C} + \bar{A}C + ADB + \cancel{A\bar{C}D}$

$\Rightarrow \bar{B}\bar{C} + \bar{A}C + ABD$ &

c) $\bar{A}\bar{C} + BC + A\bar{B} + \bar{A}B\bar{C} + \cancel{A\bar{C}\bar{D}} + \cancel{\bar{B}\bar{C}\bar{D}}$

$\Rightarrow \bar{A}\bar{C}(1+B) + BC + A\bar{B}$

$\Rightarrow \bar{A}\bar{C} + BC + A\bar{B}$

d) $\bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4 + \bar{x}_1\bar{x}_2x_4 + x_1x_2\bar{x}_3x_4 + x_2$

$\Rightarrow x_2(1 + x_1\bar{x}_3x_4) + \bar{x}_1\bar{x}_2x_4 + x_3x_4 + \bar{x}_1\bar{x}_3\bar{x}_4$

$\Rightarrow x_2 + \bar{x}_1x_4 + x_3x_4 + \bar{x}_1\bar{x}_3\bar{x}_4$

$\Rightarrow x_4x_3 + x_2 + \bar{x}_1(x_4 + \bar{x}_3\bar{x}_4) = \bar{x}_1(\bar{x}_3 + x_4) + x_2 + x_3x_4$

$\therefore \underline{\bar{x}_1\bar{x}_3 + x_3x_4 + x_2}$ ($\because \bar{x}_1x_4$ neglect)

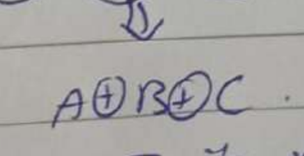
3) Eq^n: $Z = \overline{(\overline{A\bar{B}\bar{C}})(\overline{\bar{A}BC})(\overline{\bar{A}B\bar{C}})(\overline{ABC})}$

$\therefore \overline{\overline{A\bar{B}C}} + \overline{\overline{\bar{A}\bar{B}C}} + \overline{\overline{\bar{A}B\bar{C}}} + \overline{\overline{\bar{A}BC}}$

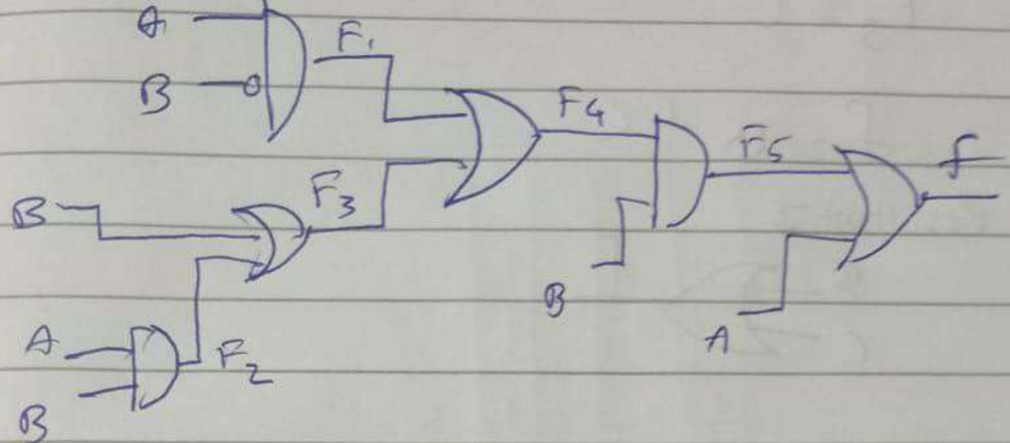
$\Rightarrow \underbrace{\bar{A}B\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C + ABC}_{\Downarrow}$

$A \oplus B \oplus C$.

$\therefore$ We use 3 i/p XOR gate.



b)



$$F_1 = A\bar{B} \quad F_2 = AB \quad F_3 = B$$

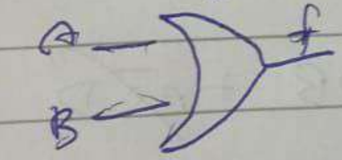
$$F_4 = A + B$$

$$F_5 = AB + B = B$$

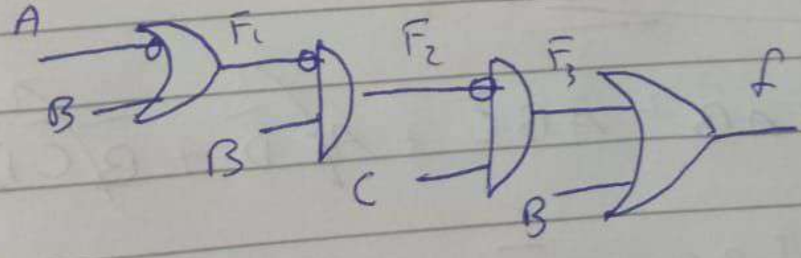
$$F_6 = A + B$$

Resultant ⇓

∴



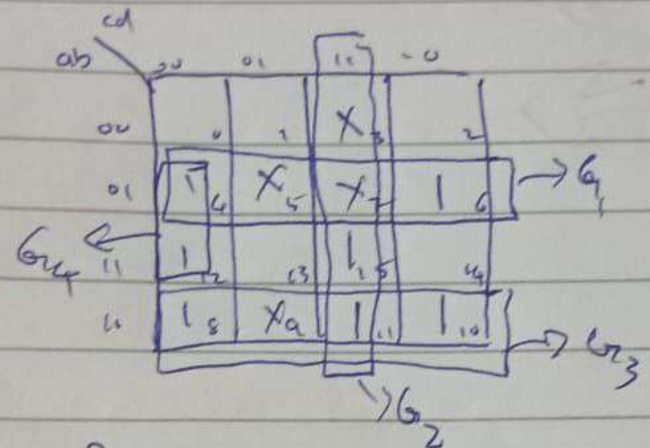
c)



$$F_1 = \bar{A} + B \quad F_2 = \overline{(\bar{A} + B)}\, B$$

$$\Rightarrow 0 \quad (\because B \cdot \bar{B} = 0)$$

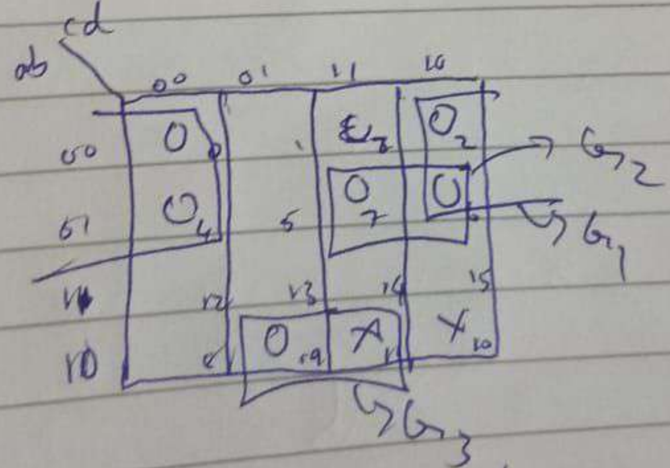
4a) $f(a,b,c,d) = \Sigma(4,6,8,10,11,12,15) + d(3,5,7,9)$



$f = G_1 + G_2 + G_3 + G_4$

$= \bar{a}b + cd + a\bar{b} + a\bar{c}\bar{d}$

b) $f(a,b,c,d) = \Pi(0,2,4,6,7,9) \cdot d(10,11)$



$f = G_1 + G_2 + G_3$

$f = (a+d)(a+b'+c')(a'+b+d')$